

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE																
Summer Examination – 2023																
Course: B. Tech.	Branch : Electrical Engineering	Semester : VI														
Subject Code & Name: Power Plant Engineering BTEEOE605B																
Max Marks: 60	Date:21/07/23	Duration: 3 Hr.														
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.																
	(Level/CO)	Marks														
Q. 1	Solve Any Two of the following.	12														
A)	Explain the terms ; i) connected Load (ii) Maximum Demand (iii) Demand Factor (iv) Diversity Factor (v) Plant Capacity Factor (vi) Plant Use Factor	CO1 6														
B)	Explain the Conventional and Non-Conventional sources of Energy	CO1 6														
C)	generating station has the following daily load cycle : <table border="1"><tr><td>Time (Hours)</td><td>0—6</td><td>6—10</td><td>10—12</td><td>12—16</td><td>16—20</td><td>20—24</td></tr><tr><td>Load (MW)</td><td>40</td><td>50</td><td>60</td><td>50</td><td>70</td><td>40</td></tr></table> Draw the load curve and find (i) maximum demand (ii) units generated per day (iii) average load and (iv) load factor.	Time (Hours)	0—6	6—10	10—12	12—16	16—20	20—24	Load (MW)	40	50	60	50	70	40	CO1 6
Time (Hours)	0—6	6—10	10—12	12—16	16—20	20—24										
Load (MW)	40	50	60	50	70	40										
Q.2	Solve Any Two of the following.	12														
A)	Describe the Classification of Turbines used in Hydro-electric power plant.	CO2 6														
B)	Explain the operational circuits of thermal power plant with Schematic diagram of thermal power plant.	CO2 6														
C)	Elaborate Pumped Storage Plant.	CO2 6														
Q. 3	Solve Any Two of the following.	12														
A)	Describe the components of Gas Power Plant with its schematic diagram and also State the advantages and disadvantages of Gas Power Plant.	CO3 6														
B)	Explain the working of diesel engine power plant and draw the schematic arrangements of it.	CO3 6														
C)	Explain working principle of Nuclear power plant with diagram also state the advantages.	CO3 6														

Q.4	Solve Any Two of the following.		12
A)	Explain Solar power generation with neat diagram of solar power plant.	CO4	6
B)	Describe the Biogas power plant.	CO4	6
C)	Explain working of MHD power plant with neat diagram.	CO4	6
Q. 5	Solve Any Two of the following.		12
A)	How will you select the choice of size and number of generator units, Explain with example.	CO5	6
B)	Explain different types of tariff.	CO5	6
C)	A consumer has a maximum demand of 200 kW at 40% load factor. If the tariff is Rs. 100 per kW of maximum demand plus 10 paise per kWh, find the overall cost per kWh.	CO5	6
	*** End ***		

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